

“BUT MAN IS PART OF NATURE, AND HIS WAR AGAINST NATURE IS INEVITABLY A WAR AGAINST HIMSELF.”

Rachel Carson, marine biologist,
from *Silent Spring*, 1962

Surtsey emerged in a volcanic eruption in 1963. Subsequent eruptions over the next four years built up the island, which—in spite of erosion—boasts more than 60 plant species.

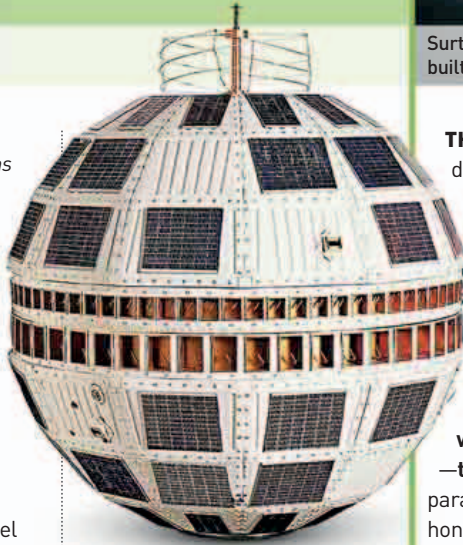
Telstar

The world's first communications satellite, Telstar transmitted its signals intermittently from 1962 to early 1963. Although it has ceased to communicate, it remains in orbit to this day.

American biologist **Gerald Edelman** (b.1929) and English biochemist **Rodney Porter** (1917–85) independently made a breakthrough that would eventually win them the Nobel Prize. They were working on **antibodies**—natural secretions that help the immune system fight infection by targeting and “neutralizing” harmful foreign particles called antigens. Edelman and Porter analyzed antibodies by chemically splitting

The structure of antibodies

An antibody molecule is made up of two “light” and two “heavy” protein chains, held together by tight bonds in a Y-shaped structure.



them into smaller constituents. They found that each Y-shaped **antibody molecule was made up of protein chains**. Their work helped unravel the chemical structure of antibodies. Later work would show how the human body produces different kinds of antibodies to target different kinds of antigens, so that the immune system can attack specific infections.

THE BELL COMPANY

developed the **first push-button telephone for public use**. Scientists also saw breakthroughs in science on a grander and more fundamental scale.

Soviet cosmonaut **Valentina Tereshkova** (b.1937) became the **first woman**—and the first civilian—to fly into space. An amateur parachutist, she became an honorary inductee into the Soviet Air Force before she trained to pilot Vostok 6. The mission helped Russian scientists understand how the female body reacted to time in space.

Scientists also reached a fundamental turning point in **understanding the nature of matter**. Experiments conducted in the 1950s had shown that

conventional subatomic particles, such as protons and neutrons, could explode to create even smaller fundamental entities. However, no one could be sure what these were. In 1963, American physicists **Murray Gell-Mann** (b.1929) and **George Zweig** (b.1937) independently proposed a “**quark model**” of matter, suggesting that a variety of different entities called quarks come together in combinations to make subatomic particles. Over the next few years, experiments in particle physics indicated that this quark model was essentially correct.

American mathematician **Edward Lorenz** (1917–2008) was revising the way we look at systems, such as weather, on a much larger scale. He suggested that a small, seemingly insignificant change in one place can have major repercussions in the long term. By alluding to the effect of tiny flapping wings leading to hurricane-scale devastation, he came up with an evocative name for his idea: **the butterfly effect**. With this, he laid the foundation for **chaos theory**.



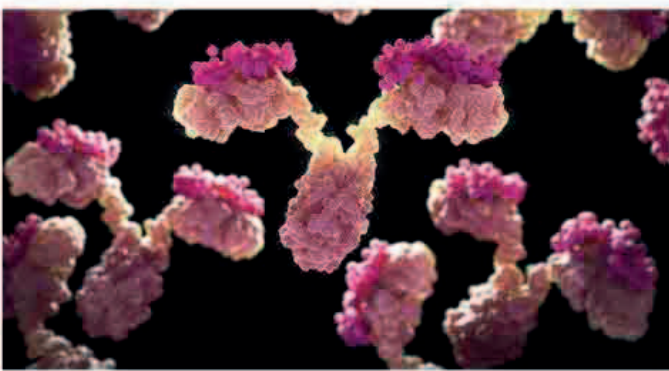
The butterfly effect

A “Lorenz attractor” is a butterfly-shaped graphical plot based on mathematical equations that describe a chaotic system.

In November, change on a massive scale was seen in the geographic realm, when an island was born near Iceland. An underwater volcano on the mid-Atlantic ridge erupted to push **Surtsey** out of the water. This gave scientists a rare chance to **study Earth's active geology** first-hand. In subsequent years, scientists were able to see how life colonized the island in a process of biological succession to form a new ecosystem.

48

THE NUMBER OF TIMES TERESHKOVA ORBITED EARTH



July 11 Telstar transmits first live transatlantic television signals from the US to Britain

British doctor **James Black** invents **beta-blockers**—a class of synthetic drugs used to treat heart disease

March 1 Thomas Starzl performs first liver transplant

June 11 James Hardy carries out first lung transplant

American physicists **Gell-Mann** and **Zweig** independently put forward the **quark theory** of subatomic particles

September 7 Fred Vine and Drummond Matthews confirm Harry Hess's theory of seafloor spreading

Dutch zoologist and animal psychologist **Niko Tinbergen** publishes method for studying animal behavior

June 16 Valentina Tereshkova becomes first woman to travel to space

Bell Company develops first push-button telephones to be made publicly available

November 14 A new island—**Surtsey**—appears above sea level following a volcanic eruption